

JSS2 Basic Science Diagnostic Assessment (Part 1)

Curriculum: Cambridge Lower Secondary (Stages 7 & 8) & Nigerian JSS2
Cluster 1: Biology - Life Processes & Living Things (Questions 1 - 20)

Science - Cluster 1: Biology

No.	Question & Options	Ans	Diagnostic Focus / Concept Tag
1	Which organelle is known as the "Powerhouse of the Cell"? A) Nucleus B) Mitochondrion C) Ribosome D) Vacuole	B	Cell structure and function (Basic).
2	In photosynthesis, which green pigment absorbs light energy? A) Hemoglobin B) Chlorophyll C) Xanthophyll D) Cytoplasm	B	Plant Biology (Essential knowledge).
3	Identify the pathway of air during inhalation: A) Nasal cavity -> Trachea -> Bronchi -> Alveoli B) Alveoli -> Bronchi -> Trachea -> Nose C) Nasal cavity -> Esophagus -> Lungs D) Mouth -> Stomach -> Alveoli	A	Human Physiology (Respiratory System).
4	Which of these describes "Respiration"? A) Breathing in and out only B) A chemical process that releases energy from glucose C) The transport of food in plants D) The process of getting rid of waste	B	Biochemical processes (Confusion check: Breathing vs Respiration).
5	An organism that makes its own food is called a/an: A) Heterotroph B) Autotroph C) Decomposer D) Parasite	B	Ecology and Nutrition.
6	The process by which plants lose water vapor through their leaves is: A) Transpiration B) Osmosis C) Respiration D) Translocation	A	Plant Physiology.
7	Which part of the blood is responsible for fighting infections? A) Red Blood Cells B) White Blood Cells C) Platelets D) Plasma	B	Human Physiology (Circulatory/Immune Support).
8	In a food chain, which organism occupies the first trophic level? A) Herbivore B) Carnivore C) Producer D) Scavenger	C	Energy Flow in Ecosystems.
9	Which of the following is a function of the Skeleton? A) Digestion B) Protection of vital organs C) Nervous coordination D) Pumping blood	B	Human Anatomy (Skeleton).
10	Which mineral is essential for the formation of healthy bones and teeth? A) Iron B) Calcium C) Iodine D) Sodium	B	Human Nutrition & Health.
11	Identify the waste product of aerobic respiration in humans: A) Oxygen B) Carbon Dioxide C) Methane D) Nitrogen	B	Metabolism waste products.
12	Which micro-organism is used in the production of bread and yoghurt? A) Virus B) Fungi (Yeast) C) Amoeba D) Plasmodium	B	Applied Biology (Micro-organisms).
13	The movement of molecules from a region of high concentration to low concentration is: A) Osmosis B) Diffusion C) Active Transport D) Digestion	B	Cellular Transport (Physical Biology).
14	Which part of the flower receives pollen during pollination? A) Anther B) Stigma C) Ovary D) Filament	B	Plant Reproduction.

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15	The enzyme in saliva that begins the digestion of starch is: A) Pepsin B) Amylase C) Lipase D) Trypsin	B	Human Physiology (Digestion).
16	Which of these is a biotic factor in an ecosystem? A) Sunlight B) Temperature C) Predation D) Rainfall	C	Environmental Biology.
17	Deficiency of Vitamin C leads to a condition called: A) Rickets B) Scurvy C) Anemia D) Beriberi	B	Deficiency Diseases.
18	What is the primary function of the large intestine? A) Digestion of proteins B) Absorption of water C) Production of bile D) Pumping air	B	Human Anatomy (Excretion/Digestion).
19	Which type of tooth is specifically sharp and used for tearing meat? A) Incisor B) Molar C) Canine D) Premolar	C	Adaptation and Nutrition.
20	Diagnostic Insight: Students failing Q4 and Q13 likely lack the conceptual foundation for Senior Secondary Biology.	-	Pathway alignment.